A SHARP RANK-r PARTIAL-TRACE INEQUALITY VIA DOUBLE-SKEW SINGULAR-VALUE BOUNDS

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Abstract. Assuming the sharp Schmidt-rank-two estimate of Fu, Gao, and Park for a double-antisymmetric projection, we prove the arbitrary-matrix rank-r partial-trace inequality for all ranks. Specifically, for every finite-dimensional bipartite Hilbert space and every operator C of rank at most r,

$$\|\operatorname{Tr}_U C\|_2^2 + \|\operatorname{Tr}_V C\|_2^2 \le r\|C\|_2^2 + \frac{1}{r}|\operatorname{Tr} C|^2.$$

The argument has two steps. The projection estimate is first converted into a two-vector action bound on the double-skew space $\mathfrak{so}(U) \otimes \mathfrak{so}(V)$. An ancilla construction and a partial transpose then reduce the rank-r inequality to pair-indexed antisymmetric modes controlled by that bound: a complete-graph summation costs only the vertex degree $r-1$, and a cancellation in one distinguished direction yields the sharp trace coefficient $1/r$. Consequences include the exact rank-constrained two-copy score curve, an exact asymmetric tensor-product score theorem and the resulting r-positivity classification, Kronecker-sum Ky Fan bounds, and Schmidt-number witnesses; none of these applications is part of the machine-checked development.

The implication from an explicit homogeneous form of the projection estimate to the displayed inequality has been machine-checked in Lean 4, together with the unconditional sharpness statements: the extremizer computation and the optimality of both coefficients. What is certified is the deduction, not the faithfulness of the formal definitions to their informal counterparts. Appendix A gives the repository, the versions, the reported axioms, and the correspondence between the formal and informal statements.

AI-generation and verification disclosure. The problem selection, prompting, and iterative direction were supplied by the author. The mathematical arguments, claimed results, applications, and exposition were generated by OpenAI's GPT-5.6 and Claude Opus 5. The Lean development of Appendix A was generated the same way, by Claude Fable 5 and Claude Opus 5 under the author's direction; it was not produced independently of the prose arguments and may share their blind spots. What a human checked there is the interface — the definitions and the conventions they encode — while Lean checks the deductions built on it. The author did not independently derive the proofs, and no claim of independent human verification is made. The manuscript has not received independent expert review and should not be relied upon until its arguments have been checked by qualified experts.

1. Introduction

Costa Rico proved the rank-sensitive partial-trace inequality below for normal operators and related its unrestricted rank-two case to two-copy distillability of Werner states [2]. Costa Rico and Wolf subsequently developed partial-trace relations for general matrices, including singular-value majorization results for Kronecker sums and applications to Schmidt-number witnesses and positive maps, while identifying the extension of the same sharp rank-sensitive bound beyond normal matrices as an open problem [3]. Fu, Gao, and Park recently resolved the rank-two endpoint through a sharp estimate for the $S(2)$-norm of a double-antisymmetric projection [4]. Their estimate supplies the sharp local rank-two input, but not by itself a mechanism for combining such information across an r-dimensional range.

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Conditional on it, the argument below supplies that mechanism and obtains the arbitrary-matrix inequality for every rank. The projection estimate is first converted into a two-vector action bound on the double-skew space $\mathfrak{so}(U) \otimes \mathfrak{so}(V)$ (Lemma 2.1). An r-dimensional ancilla recording an orthonormal basis of the range of C, a partial transpose, and a completely positive map then represent the potentially negative sector by antisymmetric modes indexed by pairs $1 \le i < j \le r$, each controlled by that bound. Grouping the modes by the vertices of the complete graph K_r, rather than estimating its $\binom{r}{2}$ edges independently, costs only the vertex degree $r-1$; a cancellation in the distinguished code direction then improves the coefficient of $|\operatorname{Tr} C|^2$ from 1 to the sharp $1/r$. Beyond threshold statements, the applications determine the exact rank-constrained score curves — the minimum of the normalized two-copy filter form over nonzero matrices of rank at most r, as in Equation (40) — for both symmetric and asymmetric two-copy filters; the asymmetric formula yields a complete r-positivity classification for the corresponding tensor-product maps.

The Werner-state and distillability terminology used in the applications follows the standard conventions [8, 5]. Throughout, $\|\cdot\|_2$ denotes the Hilbert–Schmidt, or Schatten-2, norm. The notation Tr_U means the partial trace *over* the factor U. Inner products are conjugate-linear in the first argument and linear in the second. Relative to chosen orthonormal bases, vectorization is normalized by

$$\operatorname{vec}(|x\rangle\langle y|) = x \otimes \overline{y}.$$

The symbol $\operatorname{SR}(z)$ denotes the Schmidt rank of a vector z, and $\operatorname{SN}(\sigma)$ denotes the Schmidt number of a positive operator or state σ, always across the bipartition indicated in context. With the above normalization, a vector has Schmidt rank at most k across the output–input cut exactly when it is the vectorization of a matrix of rank at most k; that is,

$$\operatorname{SR}\big(\operatorname{vec}(A)\big) = \operatorname{rank} A, \tag{1}$$

a fact used repeatedly below. Finally, for an operator on a twofold tensor power, Tr_1 denotes the partial trace *over* the first local factor, leaving an operator on the second, and Tr_2 traces over the second.

Throughout the paper we assume the following.

Hypothesis (Fu–Gao–Park). The double-antisymmetric projection estimate of [4, Theorem 2.4] holds: explicitly, Equation (5) below.

Every result stated in this paper is conditional on this hypothesis, and on nothing else. It is invoked exactly once, in the proof of Lemma 2.1; should the cited preprint be revised, the arguments here remain valid as implications from it.

Theorem 1.1 (Rank-r partial-trace inequality)**.** *Let U and V be finite-dimensional complex Hilbert spaces. If $C \in \mathcal{L}(U \otimes V)$ is nonzero and $r_0 := \operatorname{rank} C$, then*

$$\|\operatorname{Tr}_U C\|_2^2 + \|\operatorname{Tr}_V C\|_2^2 \le r_0 \|C\|_2^2 + \frac{1}{r_0} |\operatorname{Tr} C|^2. \tag{2}$$

Consequently, for every $r \in \mathbb{N}_{\ge 1}$ such that $\operatorname{rank} C \le r$*,*

$$\|\operatorname{Tr}_U C\|_2^2 + \|\operatorname{Tr}_V C\|_2^2 \le r \|C\|_2^2 + \frac{1}{r} |\operatorname{Tr} C|^2. \tag{3}$$

The assertion is immediate when $C = 0$.

Remark 1.2 (Optimality of the coefficients). The coefficients are optimal whenever $\dim U \ge r$ and $\dim V \ge 2$, or with U and V interchanged. More precisely, the coefficient r of $\|C\|_2^2$ is optimal, and conditional on choosing this optimal coefficient, the coefficient $1/r$ of $|\operatorname{Tr} C|^2$ is optimal.

Indeed, choose

$$C = P_r \otimes |v\rangle\langle w|,$$

where P_r is a rank-r projection on U and $v, w \in V$ are unit vectors. Then

$$\begin{aligned}\|\operatorname{Tr}_U C\|_2^2 &= r^2,\\ \|\operatorname{Tr}_V C\|_2^2 &= r|\langle w, v\rangle|^2,\\ \|C\|_2^2 &= r,\\ |\operatorname{Tr} C|^2 &= r^2|\langle w, v\rangle|^2.\end{aligned}$$

Thus equality holds in Equation (3), and the displayed extremizer has rank $C = r$ exactly. More generally, for nonzero C equality in the rank-r inequality requires rank $C = r$; the passage from the exact-rank inequality to a strictly larger rank parameter is strict. This is why the sharpness example is built from a rank-r projection. For a bound of the form

$$\|\operatorname{Tr}_U C\|_2^2 + \|\operatorname{Tr}_V C\|_2^2 \le a\|C\|_2^2 + b|\operatorname{Tr} C|^2,$$

taking $v \perp w$ forces $a \ge r$. Once $a = r$ is fixed, taking $v = w$ forces $b \ge 1/r$.

The proof of Theorem 1.1, and of every result it depends on, has been formalized in Lean 4, as have the statements of Remark 1.2 and Lemma 4.7. Appendix A records the repository, the toolchain and Mathlib pins, the reported axioms, and the correspondence between the formal and informal statements.

2. Double-Skew Estimates and the Marginal Operator Inequality

We first derive the double-skew action bound used below from the Fu–Gao–Park projection estimate. It is the only place where that estimate enters.

Lemma 2.1 (Double-skew action bound)**.** *Fix orthonormal bases of U and V, and let*

$$\mathfrak{so}(U) := \{L \in \mathcal{L}(U) : L^T = -L\}, \qquad \mathfrak{so}(V) := \{M \in \mathcal{L}(V) : M^T = -M\}.$$

For every K in the linear span of

$$\{L \otimes M : L \in \mathfrak{so}(U),\ M \in \mathfrak{so}(V)\},$$

and every orthonormal pair $x, y \in U \otimes V$,

$$\|Kx\|^2 + \|Ky\|^2 \le \frac{1}{2}\|K\|_2^2. \tag{4}$$

Proof. If $K = 0$, both assertions are immediate. Hence assume $K \ne 0$. Then $\dim U, \dim V \ge 2$. Let $d = \max\{\dim U, \dim V\}$. Using the orthonormal bases fixed in the statement, identify U and V with coordinate subspaces of $\mathbb{C}^d$ and extend by zero on the orthogonal complements. A block-zero extension of a skew-symmetric matrix is again skew-symmetric, and extending a finite linear combination of elementary tensors extends each tensor factor separately. Hence the extension K' of K lies in the linear span of

$$\{L \otimes M : L, M \in \mathfrak{so}(\mathbb{C}^d)\}$$

and satisfies $\|K'\|_2 = \|K\|_2$. Extending x and y by zero likewise gives an orthonormal pair x', y' with $\|K'x'\| = \|Kx\|$ and $\|K'y'\| = \|Ky\|$. It is therefore enough to prove the assertion when both local spaces have dimension d.

Label the four vectorization factors as

$$U_{\text{out}} \otimes V_{\text{out}} \otimes U_{\text{in}} \otimes V_{\text{in}},$$

where the chosen bases identify each input conjugate space with its corresponding output space. The Schmidt bipartition used below is

$$(U_{\text{out}} V_{\text{out}}) : (U_{\text{in}} V_{\text{in}}).$$

Since $L^T = -L$ implies that $\mathrm{vec}(L)$ is antisymmetric between its output and input copies, vectorization of K gives

$$Q_- \,\mathrm{vec}(K) = \mathrm{vec}(K),$$

where, on the displayed four factors,

$$Q_- := P_{\wedge^2 U}^{(U_{\mathrm{out}},U_{\mathrm{in}})} \otimes P_{\wedge^2 V}^{(V_{\mathrm{out}},V_{\mathrm{in}})}.$$

Equivalently, after the permutation that groups the two copies of U and the two copies of V, $\mathrm{vec}(K) \in (\wedge^2 U) \otimes (\wedge^2 V)$, while the Schmidt cut remains the crossed output–input cut above.

Label the four factors

$$1 = U_{\mathrm{out}}, \qquad 2 = V_{\mathrm{out}}, \qquad 3 = U_{\mathrm{in}}, \qquad 4 = V_{\mathrm{in}}.$$

Under this labeling the antisymmetric pairs $(U_{\mathrm{out}}, U_{\mathrm{in}})$ and $(V_{\mathrm{out}}, V_{\mathrm{in}})$ are Fu–Gao–Park's $(1,3)$ and $(2,4)$, while the cut $(U_{\mathrm{out}}V_{\mathrm{out}}) : (U_{\mathrm{in}}V_{\mathrm{in}})$ is their 12:34 cut. Their sharp double-antisymmetric projection estimate [4, Theorem 2.4] therefore reads: every unit vector z of Schmidt rank at most 2 across that cut satisfies

$$\langle z, Q_- z\rangle \le \frac{1}{2}. \tag{5}$$

Both sides are homogeneous of degree two in z, so for an arbitrary, not necessarily normalized, vector z of Schmidt rank at most 2,

$$\langle z, Q_- z\rangle \le \frac{1}{2}\|z\|^2. \tag{6}$$

Let $P := |x\rangle\langle x| + |y\rangle\langle y|$ be the orthogonal projection onto $\mathrm{span}\{x, y\}$, and put $A := \|KP\|_2^2$. Since $P^2 = P = P^*$,

$$\|Kx\|^2 + \|Ky\|^2 = \mathrm{Tr}(K^*KP) = A, \qquad \mathrm{rank}(KP) \le \mathrm{rank}\, P = 2. \tag{7}$$

So $\mathrm{vec}(KP)$ has Schmidt rank at most 2 across the cut above, and

$$\langle \mathrm{vec}\, K, \mathrm{vec}(KP)\rangle = \langle K, KP\rangle_{\mathrm{HS}} = A.$$

Now Q_- is a self-adjoint projection fixing $\mathrm{vec}\, K$, so Cauchy–Schwarz gives

$$A = \langle Q_- \,\mathrm{vec}\, K, \mathrm{vec}(KP)\rangle = \langle \mathrm{vec}\, K, Q_- \,\mathrm{vec}(KP)\rangle \le \|K\|_2 \, \|Q_- \,\mathrm{vec}(KP)\|,$$

while Equation (6), applied to $\mathrm{vec}(KP)$, gives

$$\|Q_- \,\mathrm{vec}(KP)\|^2 = \langle \mathrm{vec}(KP), Q_- \,\mathrm{vec}(KP)\rangle \le \frac{1}{2}A.$$

Squaring the first display and substituting the second yields $A^2 \le \frac{1}{2}A\|K\|_2^2$. If $A = 0$ the claim is trivial; otherwise divide by A to get $A \le \frac{1}{2}\|K\|_2^2$, which by Equation (7) is Equation (4). □

We next establish an operator inequality for pure states with a maximally mixed marginal.

Proposition 2.2 (Rank-r marginal operator inequality)**.** *Let $Q = \mathbb{C}^r$, let $e_1, \dots, e_r$ be an orthonormal family in $U \otimes V$, and let $q_1, \dots, q_r$ be an orthonormal basis of Q. Define the unnormalized code vector and its rank-one operator by*

$$\delta_e := \sum_{i=1}^{r} e_i \otimes q_i, \qquad \rho_0 := |\delta_e\rangle\langle\delta_e|, \qquad \Pi_e := \frac{1}{r}\rho_0.$$

Thus Π_e is the orthogonal projection onto $\mathrm{span}\{\delta_e\}$*. Write*

$$\rho^0_{UQ} := \mathrm{Tr}_V\, \rho_0, \qquad \rho^0_{VQ} := \mathrm{Tr}_U\, \rho_0.$$

Then

$$H_r^{(0)} := rI_{UVQ} + \Pi_e - \iota_{UQ}(\rho^0_{UQ}) - \iota_{VQ}(\rho^0_{VQ}) \succeq 0. \tag{8}$$

For $S \subseteq \{U, V, Q\}$, the map

$$\iota_S : \mathcal{L}\left(\bigotimes_{X \in S} X\right) \longrightarrow \mathcal{L}(U \otimes V \otimes Q)$$

uses the order inherited from U, V, Q: it tensors with identities on omitted factors and conjugates by the unique permutation unitary that restores the order $U \otimes V \otimes Q$. For example, $\iota_{UQ}(X)$ is the operator that acts as X on the labeled U, Q factors and as the identity on V.

Proof. Fix orthonormal bases of U and V, relative to which transposition and the skew-symmetric operator spaces are defined. Although the proof uses these bases, the resulting operator inequality is basis independent.

Expanding ρ_0 and transposing the UV factor gives

$$\begin{aligned} \rho_0^{T_{UV}} &= \sum_{i,j=1}^{r} \left(|e_i\rangle\langle e_j|\right)^T \otimes |q_i\rangle\langle q_j| \\ &= \sum_{i,j=1}^{r} |\bar{e}_j \otimes q_i\rangle\langle \bar{e}_i \otimes q_j|. \end{aligned} \tag{9}$$

Let

$$S := \operatorname{span}\{\bar{e}_1, \dots, \bar{e}_r\},$$

and let $J : Q \to S$ be the unitary determined by $Jq_i = \bar{e}_i$. Define a self-adjoint unitary F_J on $S \otimes Q$ by

$$F_J(s \otimes q) := Jq \otimes J^{-1}s.$$

In particular,

$$F_J(\bar{e}_i \otimes q_j) = \bar{e}_j \otimes q_i.$$

Thus the sum in Equation (9) is F_J on $S \otimes Q$. Let

$$P_\pm := \frac{1}{2}(I_{S \otimes Q} \pm F_J)$$

be its positive and negative eigenspace projections, extended by zero on $(S \otimes Q)^\perp$; thus $P_+ + P_-$ is the orthogonal projection onto $S \otimes Q$, not the identity on $U \otimes V \otimes Q$. Then

$$\rho_0^{T_{UV}} = P_+ - P_-. \tag{10}$$

Explicitly,

$$P_- = \sum_{1 \le i < j \le r} |\eta_{ij}\rangle\langle \eta_{ij}|,$$

where

$$\eta_{ij} = \frac{1}{\sqrt{2}}(\bar{e}_i \otimes q_j - \bar{e}_j \otimes q_i). \tag{11}$$

For a finite-dimensional Hilbert space E, define

$$\Lambda_E(X) := \operatorname{Tr}(X) I_E - X^T.$$

If $L_{ab} = E_{ab} - E_{ba}$, then

$$\Lambda_E(X) = \sum_{a<b} L_{ab} X L_{ab}^*. \tag{12}$$

Thus Λ_E is completely positive. Equivalently, its Choi operator is

$$J(\Lambda_E) = \sum_{i,j} |i\rangle\langle j| \otimes \Lambda_E(E_{ij}) = I - F,$$

whose spectrum is contained in $\{0, 2\}$.

Define

$$\Phi := \Lambda_U \otimes \Lambda_V \otimes \mathrm{id}_Q.$$

For an operator Y on $U \otimes V \otimes Q$, direct expansion gives

$$\begin{aligned}\Phi(Y) &= \iota_Q(\mathrm{Tr}_{UV} Y) - \iota_{VQ}\Big((\mathrm{Tr}_U Y)^{T_V}\Big) \\ &\quad - \iota_{UQ}\Big((\mathrm{Tr}_V Y)^{T_U}\Big) + Y^{T_{UV}}.\end{aligned} \tag{13}$$

Applying Equation (13) to $Y = \rho_0^{T_{UV}}$ gives

$$\Phi(\rho_0^{T_{UV}}) = I_{UVQ} - \iota_{UQ}(\rho_{UQ}^0) - \iota_{VQ}(\rho_{VQ}^0) + \rho_0. \tag{14}$$

Using Equation (10), we therefore obtain

$$H_r^{(0)} = \Phi(P_+) + \Big[(r-1)(I_{UVQ} - \Pi_e) - \Phi(P_-)\Big]. \tag{15}$$

Since $\Phi(P_+) \succeq 0$, it remains to prove

$$\Phi(P_-) \preceq (r-1)(I_{UVQ} - \Pi_e). \tag{16}$$

Let $\{L_\alpha\}_{\alpha\in\mathcal{A}_U}$ and $\{M_\beta\}_{\beta\in\mathcal{A}_V}$ be the standard skew-symmetric Kraus operators for Λ_U and Λ_V, normalized by

$$\langle L_\alpha, L_{\alpha'}\rangle_{\mathrm{HS}} = 2\delta_{\alpha\alpha'}, \qquad \langle M_\beta, M_{\beta'}\rangle_{\mathrm{HS}} = 2\delta_{\beta\beta'}.$$

Put

$$\mathcal{I}_r := \{(i,j,\alpha,\beta) : 1 \le i < j \le r,\ \alpha \in \mathcal{A}_U,\ \beta \in \mathcal{A}_V\}.$$

Define $\mathcal{T} : \ell_2(\mathcal{I}_r) \to U \otimes V \otimes Q$ by

$$\mathcal{T}c := \sum_{1\le i<j\le r} \sum_{\alpha,\beta} c_{\alpha\beta}^{(ij)} (L_\alpha \otimes M_\beta \otimes I_Q)\eta_{ij}. \tag{17}$$

Thus the column indexed by $(i,j,\alpha,\beta) \in \mathcal{I}_r$ is $(L_\alpha \otimes M_\beta \otimes I_Q)\eta_{ij}$. Using the Kraus form of Φ and the rank-one expansion of P_- therefore gives

$$\begin{aligned}\Phi(P_-) &= \sum_{i<j}\sum_{\alpha,\beta} |(L_\alpha \otimes M_\beta \otimes I_Q)\eta_{ij}\rangle\langle(L_\alpha \otimes M_\beta \otimes I_Q)\eta_{ij}| \\ &= \mathcal{T}\mathcal{T}^*.\end{aligned} \tag{18}$$

For each edge (i,j) of the complete graph K_r, define

$$K_c^{(ij)} := \sum_{\alpha,\beta} c_{\alpha\beta}^{(ij)} L_\alpha \otimes M_\beta.$$

Then $K_c^{(ij)}$ lies in the linear span of

$$\{L \otimes M : L \in \mathfrak{so}(U),\ M \in \mathfrak{so}(V)\},$$

and orthogonality of the Kraus operators gives

$$\begin{aligned}\|K_c^{(ij)}\|_2^2 &= \sum_{\alpha,\beta} |c_{\alpha\beta}^{(ij)}|^2 \|L_\alpha \otimes M_\beta\|_2^2 \\ &= 4\|c^{(ij)}\|_2^2.\end{aligned} \tag{19}$$

Grouping the output of $\mathcal{T}$ according to the basis $\{q_k\}$ gives

$$\mathcal{T}c = \frac{1}{\sqrt{2}} \sum_{k=1}^{r} \left(\sum_{i<k} K_c^{(ik)} \overline{e}_i - \sum_{k<j} K_c^{(kj)} \overline{e}_j \right) \otimes q_k. \tag{20}$$

At each vertex k, there are $r-1$ incident terms. Hence Cauchy–Schwarz yields

$$\begin{aligned}\|\mathcal{T}c\|^2 &\le \frac{r-1}{2}\sum_{k=1}^{r}\left(\sum_{i<k}\|K_c^{(ik)}\overline{e}_i\|^2+\sum_{k<j}\|K_c^{(kj)}\overline{e}_j\|^2\right)\\ &= \frac{r-1}{2}\sum_{1\le i<j\le r}\left(\|K_c^{(ij)}\overline{e}_i\|^2+\|K_c^{(ij)}\overline{e}_j\|^2\right).\end{aligned} \tag{21}$$

By Lemma 2.1, each parenthesized expression is at most $\frac{1}{2}\|K_c^{(ij)}\|_2^2$. Therefore,

$$\begin{aligned}\|\mathcal{T}c\|^2 &\le \frac{r-1}{4}\sum_{i<j}\|K_c^{(ij)}\|_2^2\\ &= (r-1)\|c\|_2^2,\end{aligned} \tag{22}$$

where the last equality follows from Equation (19). The constant bookkeeping is

$$\frac{r-1}{2}\cdot\frac{1}{2}\cdot 4 = r-1:$$

the three factors are respectively the vertex Cauchy–Schwarz bound including the $1/\sqrt{2}$ in η_{ij}, the double-skew action bound, and the Kraus normalization. This estimate is attained already when $r=2$ and $\dim U=\dim V=2$. Then every vertex has degree one and $\mathfrak{so}(2)\otimes\mathfrak{so}(2)$ is one-dimensional; its nonzero elements are scalar multiples of a unitary, so every orthonormal pair saturates Equation (4). There is no slack anywhere in that $r=2$ chain.

Thus

$$\mathcal{T}\mathcal{T}^*\preceq (r-1)I_{UVQ}.$$

It remains to use the orthogonality of the range of $\mathcal{T}$ to δ_e. Since each tensor product of two skew-symmetric operators is symmetric,

$$K^T=K \qquad \text{for every } K \text{ in the above linear span.}$$

Consequently,

$$\begin{aligned}\langle\delta_e,\mathcal{T}c\rangle &= \frac{1}{\sqrt{2}}\sum_{i<j}\left(\overline{e}_j^TK_c^{(ij)}\overline{e}_i-\overline{e}_i^TK_c^{(ij)}\overline{e}_j\right)\\ &= 0.\end{aligned} \tag{23}$$

Hence

$$\operatorname{ran}\mathcal{T}\subseteq\delta_e^\perp, \qquad P_{\delta_e^\perp}=I_{UVQ}-\Pi_e.$$

Since Equation (22) says that $\|\mathcal{T}\|^2\le r-1$, for every vector x,

$$\begin{aligned}\langle x,\mathcal{T}\mathcal{T}^*x\rangle &= \langle P_{\delta_e^\perp}x,\mathcal{T}\mathcal{T}^*P_{\delta_e^\perp}x\rangle\\ &\le (r-1)\|P_{\delta_e^\perp}x\|^2.\end{aligned}$$

Equivalently,

$$\mathcal{T}\mathcal{T}^*\preceq (r-1)(I_{UVQ}-\Pi_e).$$

This proves Equation (16), and therefore $H_r^{(0)}\succeq 0$ by Equation (15). ☐

Remark 2.3 (Transpose parity, and where the trace term comes from). The only structural input to Equation (23) is that every K in the span of $\{L\otimes M\}$ satisfies $K^T=K$, and that holds because a tensor product of an *even* number of skew-symmetric factors is symmetric. Had the number of factors been odd, one would have $K^T=-K$, the two terms of Equation (23) would add instead of cancelling, and $\operatorname{ran}\mathcal{T}$ would not lie in $\delta_e^\perp$. The argument would then yield only $\mathcal{T}\mathcal{T}^*\preceq(r-1)I$ in place of $(r-1)(I-\Pi_e)$, giving $rI+\rho_0-\iota_{UQ}(\rho_{UQ}^0)-\iota_{VQ}(\rho_{VQ}^0)\succeq 0$ and hence the coefficient 1 rather than the sharp $1/r$ in front of $|\operatorname{Tr}C|^2$.

So the $\frac{1}{r}|\operatorname{Tr} C|^2$ term of Theorem 1.1 is exactly the contribution of the single protected direction δ_e, and transpose parity is what protects it. This also indicates which generalizations can be expected to keep the sharp trace coefficient: those built from an even number of skew factors.

Remark 2.4 (The marginal inequality as an r-positivity statement). Write $\Delta(X) := \operatorname{Tr}(X)I$, let τ denote transposition, and recall $\mathcal{R}_\alpha(X) = \operatorname{Tr}(X)I + \alpha X$ from Corollary 4.6. On a product operator $X_U \otimes X_V$ one has $\tau(X_U \otimes X_V) = X_U^T \otimes X_V^T$ and therefore $\Lambda_U(X_U^T) = t_U I - X_U$ with $t_U := \operatorname{Tr} X_U$, so $(\Lambda_U \otimes \Lambda_V) \circ \tau = \mathcal{R}_{-1}^U \otimes \mathcal{R}_{-1}^V$. Adding $(r-1)(t_U t_V I - \frac{1}{r} X_U \otimes X_V)$ gives

$$rt_U t_V\, I - t_U\, I \otimes X_V - t_V\, X_U \otimes I + \tfrac{1}{r}\, X_U \otimes X_V = r\left(t_U I - \tfrac{1}{r} X_U\right) \otimes \left(t_V I - \tfrac{1}{r} X_V\right),$$

and since both sides are linear and product operators span, for every $r \geq 1$

$$(\Lambda_U \otimes \Lambda_V) \circ \tau + (r-1)\left(\Delta - \tfrac{1}{r}\mathrm{id}\right) = r\, \mathcal{R}_{-1/r}^U \otimes \mathcal{R}_{-1/r}^V. \tag{24}$$

Now $\operatorname{Tr}_{UV} \rho_0 = I_Q$ because the e_i are orthonormal, and $\rho_0 = r\Pi_e$, so $I - \Pi_e = \left[(\Delta - \frac{1}{r}\mathrm{id}) \otimes \mathrm{id}_Q\right](\rho_0)$. Combining this with Equation (14) and Equation (24) rewrites Equation (8) as

$$H_r^{(0)} = \left[(r\, \mathcal{R}_{-1/r}^U \otimes \mathcal{R}_{-1/r}^V) \otimes \mathrm{id}_Q\right](\rho_0).$$

Thus Proposition 2.2 states precisely that $\mathcal{R}_{-1/r} \otimes \mathcal{R}_{-1/r}$ is r-positive; conjugating ρ_0 by a filter on the ancilla passes from orthonormal frames $e_1, \ldots, e_r$ to arbitrary vectors of Schmidt rank at most r. Equivalently, it is the sufficiency half of Corollary 4.9 at $a = b = 1/r$, which is thus available already here, before Theorem 1.1.

3. Proof of the Rank-r Partial-Trace Inequality

Lemma 3.1 (Partial-trace contraction identities)**.** *Let $e_1, \ldots, e_s \in U \otimes V$ be orthonormal, let $d_1, \ldots, d_s \in U \otimes V$, and let $q_1, \ldots, q_s$ be an orthonormal basis of $Q = \mathbb{C}^s$. Define*

$$C := \sum_{i=1}^{s} |e_i\rangle\langle d_i|, \qquad \delta_e := \sum_{i=1}^{s} e_i \otimes q_i, \qquad \rho_0 := |\delta_e\rangle\langle\delta_e|,$$

and

$$\delta_d := \sum_{i=1}^{s} d_i \otimes q_i, \qquad \rho_{UQ}^0 := \operatorname{Tr}_V \rho_0, \qquad \rho_{VQ}^0 := \operatorname{Tr}_U \rho_0.$$

Then

$$\langle \delta_d, \delta_d \rangle = \|C\|_2^2, \tag{25}$$

$$\langle \delta_d, \rho_0 \delta_d \rangle = |\operatorname{Tr} C|^2, \tag{26}$$

$$\langle \delta_d, \iota_{UQ}(\rho_{UQ}^0)\delta_d \rangle = \|\operatorname{Tr}_U C\|_2^2, \tag{27}$$

$$\langle \delta_d, \iota_{VQ}(\rho_{VQ}^0)\delta_d \rangle = \|\operatorname{Tr}_V C\|_2^2. \tag{28}$$

Proof. The first identity follows from the orthonormality of the e_i:

$$\|C\|_2^2 = \sum_i \|d_i\|^2 = \langle \delta_d, \delta_d \rangle.$$

Since $\operatorname{Tr} C = \sum_i \langle d_i, e_i \rangle$ under the stated inner-product convention,

$$\langle \delta_e, \delta_d \rangle = \sum_i \langle e_i, d_i \rangle = \overline{\operatorname{Tr} C},$$

which proves Equation (26).

For the first marginal contraction,

$$\rho_{UQ}^0 = \sum_{i,j} \operatorname{Tr}_V |e_i\rangle\langle e_j| \otimes |q_i\rangle\langle q_j|,$$

and hence

$$\langle \delta_d, \iota_{UQ}(\rho^0_{UQ})\delta_d\rangle = \sum_{i,j} \langle d_i, (\mathrm{Tr}_V |e_i\rangle\langle e_j| \otimes I_V) d_j\rangle. \tag{29}$$

Choose product bases $\{u_a\}$ of U and $\{v_b\}$ of V, and write

$$e_i = \sum_{a,b} e_{i,ab} u_a \otimes v_b, \qquad d_i = \sum_{a,b} d_{i,ab} u_a \otimes v_b.$$

Then

$$(\mathrm{Tr}_U C)_{b,c} = \sum_{i,a} e_{i,ab} \overline{d_{i,ac}},$$

so

$$\|\mathrm{Tr}_U C\|_2^2 = \sum_{i,j} \sum_{a,a'} \sum_{b,c} e_{i,ab} \overline{d_{i,ac}}\, \overline{e_{j,a'b}} d_{j,a'c}. \tag{30}$$

Expanding the right-hand side of Equation (29) gives, explicitly,

$$\begin{aligned}\sum_{i,j} &\langle d_i, (\mathrm{Tr}_V |e_i\rangle\langle e_j| \otimes I_V) d_j\rangle \\ &= \sum_{i,j} \sum_{a,a'} \sum_{b,c} e_{i,ab} \overline{d_{i,ac}}\, \overline{e_{j,a'b}} d_{j,a'c},\end{aligned}$$

the same quadruple sum as in Equation (30). This proves Equation (27). Interchanging U and V proves Equation (28). ☐

Proof of Theorem 1.1. The case $C = 0$ is immediate. Suppose $C \neq 0$, and let

$$r_0 := \operatorname{rank} C.$$

Choose an orthonormal basis $\{e_i\}_{i=1}^{r_0}$ of $\operatorname{ran} C$ and set

$$d_i := C^* e_i.$$

Then

$$C = \sum_{i=1}^{r_0} |e_i\rangle\langle d_i|, \qquad \sum_{i=1}^{r_0} \|d_i\|^2 = \|C\|_2^2. \tag{31}$$

Since $\operatorname{rank} C \leq r$, one has $1 \leq r_0 \leq r$.

Let $Q = \mathbb{C}^{r_0}$, with orthonormal basis $\{q_i\}_{i=1}^{r_0}$, and define

$$\delta_e := \sum_{i=1}^{r_0} e_i \otimes q_i, \qquad \rho_0 := |\delta_e\rangle\langle \delta_e|, \qquad \Pi_e := \frac{1}{r_0}\rho_0, \qquad \delta_d := \sum_{i=1}^{r_0} d_i \otimes q_i.$$

Write $\rho^0_{UQ} = \mathrm{Tr}_V \rho_0$ and $\rho^0_{VQ} = \mathrm{Tr}_U \rho_0$. Applying Proposition 2.2 with $r = r_0$ gives $H^{(0)}_{r_0} \succeq 0$. Therefore,

$$\begin{aligned}0 &\leq \langle \delta_d, H^{(0)}_{r_0} \delta_d\rangle \\ &= r_0 \langle \delta_d, \delta_d\rangle + \frac{1}{r_0}\langle \delta_d, \rho_0 \delta_d\rangle - \langle \delta_d, \iota_{UQ}(\rho^0_{UQ})\delta_d\rangle - \langle \delta_d, \iota_{VQ}(\rho^0_{VQ})\delta_d\rangle.\end{aligned} \tag{32}$$

By Lemma 3.1, this is precisely

$$\|\mathrm{Tr}_U C\|_2^2 + \|\mathrm{Tr}_V C\|_2^2 \leq r_0 \|C\|_2^2 + \frac{1}{r_0} |\mathrm{Tr}\, C|^2. \tag{33}$$

It remains to pass from r_0 to r. From Equation (31) and $\operatorname{Tr} C = \sum_i \langle d_i, e_i \rangle$,

$$\begin{aligned} |\operatorname{Tr} C|^2 &\le \left(\sum_{i=1}^{r_0} \|d_i\| \right)^2 \\ &\le r_0 \sum_{i=1}^{r_0} \|d_i\|^2 = r_0 \|C\|_2^2. \end{aligned}$$

The passage from r_0 to r is then a single identity: for all $r \ge r_0 \ge 1$,

$$r\|C\|_2^2 + \frac{1}{r}|\operatorname{Tr} C|^2 - \left(r_0\|C\|_2^2 + \frac{1}{r_0}|\operatorname{Tr} C|^2 \right) = (r - r_0) \left(\|C\|_2^2 - \frac{|\operatorname{Tr} C|^2}{r r_0} \right). \tag{34}$$

By the trace–rank bound the second factor is at least $\left(1 - \frac{1}{r}\right)\|C\|_2^2 \ge 0$, and the first is nonnegative, so the right-hand side is nonnegative. Combining this with Equation (33) proves Equation (3).

Both factors are moreover *strictly* positive when $C \neq 0$ and $r_0 < r$: the first is at least 1, and $r_0 < r$ forces $r \ge 2$, so $1 - \frac{1}{r} \ge \frac{1}{2}$ while $\|C\|_2^2 > 0$. Hence the inequality of Equation (3) is strict whenever $\operatorname{rank} C < r$ and $C \neq 0$, which is the exact-rank assertion of Remark 1.2. □

Remark 3.2 (Dependence on the constant of Lemma 2.1). Every constant in Sections 2 and 3 is affine in the constant of Equation (4), so what the argument proves is really a one-parameter family. Suppose Equation (4) holds with $\frac{1}{2}$ replaced by some c, that is, $\|Kx\|^2 + \|Ky\|^2 \le c\|K\|_2^2$. Tracking c through Equations (21) and (22) turns Equation (16) into $\Phi(P_-) \preceq 2c(r-1)(I - \Pi_e)$, whence

$$H_r^{(0)} + (2c - 1)(r - 1)(I - \Pi_e) \succeq 0,$$

and contracting against δ_d as in the proof above gives, for nonzero C with $r = \operatorname{rank} C$,

$$\|\operatorname{Tr}_U C\|_2^2 + \|\operatorname{Tr}_V C\|_2^2 \le \left[r + (2c-1)(r-1)\right]\|C\|_2^2 + \left[1 - (2c-1)(r-1)\right]\frac{1}{r}|\operatorname{Tr} C|^2. \tag{35}$$

At $c = \frac{1}{2}$ this is Equation (2).

The value $\frac{1}{2}$ cannot be improved: as noted after Equation (22) it is already attained at $\dim U = \dim V = 2$, where $\mathfrak{so}(2) \otimes \mathfrak{so}(2)$ is one-dimensional and every orthonormal pair saturates Equation (4). So necessarily $c \ge \frac{1}{2}$, and Equation (35) is never stronger than Theorem 1.1. Its point is the opposite one. The standing hypothesis is a single recent preprint; Equation (35) records that the deduction from it is stable, in that if the estimate of [4] should turn out to hold only with some larger c, the conclusion here degrades continuously in c instead of failing outright.

4. Applications and Corollaries

This section collects consequences of the results proved above: the exact symmetric score curve, an asymmetric tensor-product score theorem, block positivity and positive maps, quantitative Schmidt-number separation, Ky Fan bounds for Kronecker sums, and the exact Schmidt number of a product of antisymmetric projector states. None of them is part of the machine-checked development.

They do not all rest on the same input. Proposition 4.1, Theorem 4.8, and Corollary 4.11 use Theorem 1.1 at general r, and are the results for which the extension beyond normal matrices is what is needed; Corollaries 4.5, 4.6 and 4.9 restate them through the Choi–Jamiołkowski correspondence. Corollaries 4.12 and 4.13, by contrast, rest on Lemma 2.1 alone – that is, on the Fu–Gao–Park estimate alone – and do not use Theorem 1.1. Appendix C collects two further consequences that use nothing about the witnesses constructed here beyond r-block positivity.

4.1. **Two-copy block positivity.** Fix $d \geq 2$ and $\alpha \in \mathbb{R}$. Let

$$\rho_\alpha := I + \alpha F$$

be the unnormalized Werner operator on $\mathbb{C}^d \otimes \mathbb{C}^d$; for $\alpha \in [-1, 1]$, it is proportional to the standard Werner state [8]. Its partial transpose is

$$\rho_\alpha^\Gamma = I + \alpha|\Omega\rangle\langle\Omega|, \qquad \Omega := \sum_{i=1}^{d} e_i \otimes e_i.$$

The case $\alpha \geq 0$ is immediate because

$$I + \alpha|\Omega\rangle\langle\Omega| \succeq 0.$$

For $\alpha \leq 0$ and $C \in M_{d^2}(\mathbb{C})$, define

$$q^{(2)}(\alpha, C) := \|C\|_2^2 + \alpha\left(\|\mathrm{Tr}_1\, C\|_2^2 + \|\mathrm{Tr}_2\, C\|_2^2\right) + \alpha^2|\mathrm{Tr}\, C|^2. \tag{36}$$

The first argument is the Werner parameter of ρ_α; only $\alpha \leq 0$ is at issue, and it is convenient to write $\alpha = -t$ with $t \geq 0$, so that

$$q^{(2)}(-t, C) = \|C\|_2^2 - t\left(\|\mathrm{Tr}_1\, C\|_2^2 + \|\mathrm{Tr}_2\, C\|_2^2\right) + t^2|\mathrm{Tr}\, C|^2.$$

An operator is called r-block-positive if its expectation is nonnegative on every vector of Schmidt rank at most r. With the vectorization convention fixed in the introduction and the regrouping $(A_1A_2) : (B_1B_2)$, one has

$$\begin{aligned}&\left\langle \mathrm{vec}(C), (\rho_\alpha^\Gamma)^{\otimes 2}\, \mathrm{vec}(C)\right\rangle \\ &= q^{(2)}(\alpha, C),\end{aligned} \tag{37}$$

which is what the first argument of $q^{(2)}$ records. Moreover,

$$\mathrm{SR}\big(\mathrm{vec}(C)\big) = \mathrm{rank}\, C$$

across the same output–input regrouping. Thus r-block positivity of $(\rho_{-t}^\Gamma)^{\otimes 2}$ is equivalent to $q^{(2)}(-t, C) \geq 0$ for every matrix C of rank at most r. For every C of rank at most r, Equation (3) implies

$$q^{(2)}(-t, C) \geq (1 - rt)\|C\|_2^2 + \left(t^2 - \frac{t}{r}\right)|\mathrm{Tr}\, C|^2. \tag{38}$$

If $0 \leq t \leq 1/r$, then $t^2 - t/r \leq 0$. Using

$$|\mathrm{Tr}\, C|^2 \leq r\|C\|_2^2$$

in the appropriate direction gives

$$\begin{aligned} q^{(2)}(-t, C) &\geq \left(1 - rt + rt^2 - t\right)\|C\|_2^2 \\ &= (1 - t)(1 - rt)\|C\|_2^2 \geq 0. \end{aligned} \tag{39}$$

Proposition 4.1 (Exact rank-constrained two-copy score curve)**.** *Let $d \geq 2$ and $1 \leq r \leq d$. For $t \geq 0$, define*

$$\mu_r(t) := \inf_{\substack{C \in M_{d^2}(\mathbb{C})\setminus\{0\} \\ \mathrm{rank}\, C \leq r}} \frac{q^{(2)}(-t, C)}{\|C\|_2^2}.$$

Then

$$\mu_r(t) = \begin{cases} (1 - t)(1 - rt), & 0 \leq t \leq 1/r, \\ 1 - rt, & t \geq 1/r. \end{cases} \tag{40}$$

Both branches are attained by rank-r matrices.

Proof. For $C \neq 0$ of rank at most r, set

$$z := \frac{|\operatorname{Tr} C|^2}{\|C\|_2^2}.$$

The trace–rank estimate gives $0 \leq z \leq r$. Dividing Equation (38) by $\|C\|_2^2$ yields

$$\frac{q^{(2)}(-t, C)}{\|C\|_2^2} \geq 1 - rt + \left(t^2 - \frac{t}{r}\right) z. \tag{41}$$

If $0 \leq t \leq 1/r$, the coefficient of z is nonpositive, so Equation (41) and $z \leq r$ give

$$\frac{q^{(2)}(-t, C)}{\|C\|_2^2} \geq 1 - rt + r\left(t^2 - \frac{t}{r}\right) = (1-t)(1-rt).$$

Choose a rank-r projection $P_r \in M_d(\mathbb{C})$ and a unit vector $v \in \mathbb{C}^d$. For

$$C = P_r \otimes |v\rangle\langle v|$$

one has

$$\|C\|_2^2 = r, \qquad |\operatorname{Tr} C|^2 = r^2,$$
$$\|\operatorname{Tr}_1 C\|_2^2 = r^2, \quad \|\operatorname{Tr}_2 C\|_2^2 = r,$$

and hence equality holds.

If $t \geq 1/r$, the coefficient of z in Equation (41) is nonnegative, so

$$\frac{q^{(2)}(-t, C)}{\|C\|_2^2} \geq 1 - rt.$$

Choose orthogonal unit vectors $v, w \in \mathbb{C}^d$ and set

$$C = P_r \otimes |v\rangle\langle w|.$$

Then

$$\operatorname{Tr} C = 0, \qquad \|\operatorname{Tr}_1 C\|_2^2 = r^2, \qquad \operatorname{Tr}_2 C = 0,$$

so equality again holds. □

Remark 4.2 (Why the range $r \leq d$ is essential). The extremizers in Proposition 4.1 require a rank-r projection in one d-dimensional tensor factor. The score formula does not in general extend to $r > d$ by replacing r with $\min\{r, d\}$. For example, when the rank bound is $r = d^2$, no rank constraint remains. If $0 \leq t \leq 1/d$, the smallest eigenvalue of

$$\left(\rho_{-t}^{\Gamma}\right)^{\otimes 2}$$

is

$$(1 - dt)^2.$$

By Equation (37), the displayed infimum is the smallest eigenvalue, and therefore

$$\inf_{C \neq 0} \frac{q^{(2)}(-t, C)}{\|C\|_2^2} = (1 - dt)^2.$$

This generally differs from $(1-t)(1-dt)$, the expression obtained by formally substituting d for r in the first branch of Equation (40).

Corollary 4.3 (An explicit adjacent-Schmidt-rank violation)**.** *Let $1 \leq r < d$ and put*

$$Z_{r,d} := \left(I - \frac{1}{r}|\Omega\rangle\langle\Omega|\right)^{\otimes 2}.$$

Across the regrouped bipartition $(A_1A_2) : (B_1B_2)$,

$$\langle \psi, Z_{r,d}\psi \rangle \geq 0 \qquad \textit{whenever } \|\psi\| = 1 \textit{ and } \operatorname{SR}(\psi) \leq r.$$

On the other hand, there is a unit vector ψ_{r+1} of Schmidt rank $r+1$ such that

$$\langle \psi_{r+1}, Z_{r,d}\psi_{r+1}\rangle = -\frac{1}{r}. \tag{42}$$

Proof. The nonnegativity assertion is Corollary 4.5 at $t = 1/r$. Choose a rank-$(r+1)$ projection P_{r+1} and orthogonal unit vectors $v, w \in \mathbb{C}^d$, and define

$$C = P_{r+1} \otimes |v\rangle\langle w|, \qquad \psi_{r+1} := \frac{\mathrm{vec}(C)}{\sqrt{r+1}}.$$

Then $\mathrm{SR}(\psi_{r+1}) = \operatorname{rank} C = r+1$, while

$$\frac{q^{(2)}(-1/r, C)}{\|C\|_2^2} = 1 - \frac{r+1}{r} = -\frac{1}{r}.$$

Now use Equation (37). ☐

Remark 4.4 (A strict signed separation). Let $0 < \varepsilon < 1/(r+1)$ and set $t = (1-\varepsilon)/r$. Then Proposition 4.1 gives

$$\frac{q^{(2)}(-t, C)}{\|C\|_2^2} \geq \varepsilon\left(1 - \frac{1-\varepsilon}{r}\right) > 0 \qquad (\operatorname{rank} C \leq r),$$

whereas the rank-$(r+1)$ matrix used in Corollary 4.3 has normalized score

$$1 - (r+1)t = \frac{(r+1)\varepsilon - 1}{r} < 0.$$

Thus the filter has a uniform positive margin on all normalized matrices of rank at most r, while the displayed rank-$(r+1)$ matrix has negative score.

This yields the exact r-block-positivity threshold for the two-copy family $(\rho_\alpha^\Gamma)^{\otimes 2}$.

Corollary 4.5 (Two-copy block positivity)**.** *Let $d \geq 2$ and $\alpha \in \mathbb{R}$. For every $1 \leq r \leq d^2$, the operator*

$$(\rho_\alpha^\Gamma)^{\otimes 2}$$

is r-block-positive across the regrouped bipartition

$$(A_1A_2) : (B_1B_2)$$

if and only if

$$\alpha \geq -\frac{1}{\min\{r, d\}}. \tag{43}$$

Proof. It remains only to consider $\alpha = -t < 0$.

Suppose first that $r \leq d$. If $t \leq 1/r$, then Equation (39) gives

$$q^{(2)}(-t, C) \geq 0$$

for every C of rank at most r, proving r-block positivity.

For necessity in all cases, set

$$s := \min\{r, d\}.$$

Choose a rank-s projection P_s and orthogonal unit vectors $v, w \in \mathbb{C}^d$, and let

$$C = P_s \otimes |v\rangle\langle w|.$$

Then

$$\operatorname{rank} C = s \le r,$$
$$\operatorname{Tr} C = 0,$$
$$\|C\|_2^2 = s,$$
$$\|\operatorname{Tr}_1 C\|_2^2 = s^2,$$
$$\|\operatorname{Tr}_2 C\|_2^2 = 0.$$

Hence

$$q^{(2)}(-t, C) = s(1 - st),$$

which is negative whenever $t > 1/s$. Thus r-block positivity forces

$$t \le \frac{1}{\min\{r, d\}}.$$

Finally, suppose $r > d$. If $t \le 1/d$, then the Choi operator of

$$\mathcal{R}_{-t}(X) := \operatorname{Tr}(X)I - tX$$

is

$$J(\mathcal{R}_{-t}) = I - t|\Omega\rangle\langle\Omega| \succeq 0,$$

because $\|\Omega\|^2 = d$ and the only potentially nonpositive eigenvalue is $1 - td$. Therefore $\mathcal{R}_{-t}$ is completely positive, so $\mathcal{R}_{-t}^{\otimes 2}$ is completely positive and hence r-positive for every r. This proves sufficiency when $r > d$. □

4.2. **r-positivity of the tensor square.** The preceding block-positivity statement has an equivalent map formulation through the Choi–Jamiołkowski correspondence: a map is r-positive if and only if its Choi operator is r-block-positive [1, 7]. With the output factors grouped against the input factors, the Choi operator of $\mathcal{R}_\alpha^{\otimes 2}$ is exactly $(\rho_\alpha^\Gamma)^{\otimes 2}$ across the bipartition $(A_1A_2) : (B_1B_2)$ used above.

Corollary 4.6 (Tensor-square stability)**.** *Let $d \ge 2$ and $\alpha \in \mathbb{R}$. Define*

$$\mathcal{R}_\alpha(X) := \operatorname{Tr}(X)I + \alpha X.$$

For every $1 \le r \le d^2$, the tensor-square map

$$\mathcal{R}_\alpha^{\otimes 2}$$

is r-positive if and only if

$$\alpha \ge -\frac{1}{\min\{r, d\}}.$$

4.3. **Exact asymmetric tensor-product theorem.** For $a, b \ge 0$, define

$$q_{a,b}(C) := \|C\|_2^2 - a\|\operatorname{Tr}_1 C\|_2^2 - b\|\operatorname{Tr}_2 C\|_2^2 + ab|\operatorname{Tr} C|^2. \tag{44}$$

Under the same vectorization and regrouping as above,

$$q_{a,b}(C) = \langle \operatorname{vec}(C), (I - a|\Omega\rangle\langle\Omega|) \otimes (I - b|\Omega\rangle\langle\Omega|) \operatorname{vec}(C)\rangle .$$

Lemma 4.7 (One-sided rank bound)**.** *If $\operatorname{rank} C \le r$, then*

$$\|\operatorname{Tr}_j C\|_2^2 \le r\|C\|_2^2, \qquad j \in \{1, 2\}. \tag{45}$$

Proof. Write $C = \sum_{i=1}^{r_0} |e_i\rangle\langle d_i|$ as in Equation (31), with $e_1, \ldots, e_{r_0}$ orthonormal and $r_0 = \operatorname{rank} C \le r$. Identify a vector x of the bipartite space with its coefficient matrix X relative to the chosen product basis, so that $\|X\|_2 = \|x\|$. The two partial traces of a rank-one operator are then matrix products,

$$\operatorname{Tr}_2|x\rangle\langle y| = XY^*, \qquad \operatorname{Tr}_1|x\rangle\langle y| = \overline{X^*Y}, \tag{46}$$

both immediate from the definitions. The Hilbert–Schmidt norm is submultiplicative, so each summand obeys $\|\mathrm{Tr}_j|e_i\rangle\langle d_i|\|_2 \le \|e_i\|\|d_i\| = \|d_i\|$. By the triangle inequality and then Cauchy–Schwarz,

$$\|\mathrm{Tr}_j C\|_2^2 \le \Big(\sum_{i=1}^{r_0}\|d_i\|\Big)^2 \le r_0\sum_{i=1}^{r_0}\|d_i\|^2 = r_0\|C\|_2^2 \le r\|C\|_2^2,$$

the equality being Equation (31). ☐

Recall that Tr_1 means the trace *over* the first local factor, leaving an operator on the second, while Tr_2 traces over the second and leaves an operator on the first. Thus the coefficient a in Equation (44) multiplies the first-factor trace and b multiplies the second-factor trace.

Theorem 4.8 (Exact asymmetric tensor-product score)**.** *Let $d \ge 2$, $1 \le r \le d$, and $a, b \ge 0$. Put*

$$m := \max\{a,b\}, \qquad \ell := \min\{a,b\}.$$

Then

$$\inf_{\substack{C\in M_{d^2}(\mathbb{C})\setminus\{0\}\\ \mathrm{rank}\, C\le r}} \frac{q_{a,b}(C)}{\|C\|_2^2} = \begin{cases}(1-\ell)(1-rm), & m \le 1/r,\\ 1-rm, & m \ge 1/r.\end{cases} \tag{47}$$

Proof. By interchanging the two tensor factors, assume $a = m$ and $b = \ell$. For $C \ne 0$ of rank at most r, write

$$x := \frac{\|\mathrm{Tr}_1 C\|_2^2}{\|C\|_2^2}, \qquad y := \frac{\|\mathrm{Tr}_2 C\|_2^2}{\|C\|_2^2}, \qquad z := \frac{|\mathrm{Tr}\, C|^2}{\|C\|_2^2}.$$

By Theorem 1.1 and Lemma 4.7,

$$x + y \le r + \frac{z}{r}, \qquad x \le r, \qquad 0 \le z \le r.$$

Consequently,

$$\begin{aligned} ax + by &= b(x+y) + (a-b)x \\ &\le b\left(r + \frac{z}{r}\right) + (a-b)r = ar + \frac{b}{r}z, \end{aligned}$$

and hence

$$\frac{q_{a,b}(C)}{\|C\|_2^2} \ge 1 - ar + b\left(a - \frac{1}{r}\right)z. \tag{48}$$

If $a \le 1/r$, minimize the right-hand side over $0 \le z \le r$ by taking $z = r$; this gives

$$1 - ar + br\left(a - \frac{1}{r}\right) = (1-b)(1-ar).$$

Equality is attained by $C = P_r \otimes |v\rangle\langle v|$. If $a \ge 1/r$, minimize by taking $z = 0$; this gives $1 - ar$, with equality attained by $C = P_r \otimes |v\rangle\langle w|$ for $v \perp w$. If $b > a$, interchange the two tensor factors in these constructions. ☐

Corollary 4.9 (Asymmetric tensor-product r-positivity threshold)**.** *Let $d \ge 2$, $1 \le r \le d^2$, and $a, b \ge 0$. Then*

$$(I - a|\Omega\rangle\langle\Omega|) \otimes (I - b|\Omega\rangle\langle\Omega|)$$

is r-block-positive across the regrouped bipartition if and only if

$$\max\{a,b\} \le \frac{1}{\min\{r,d\}}. \tag{49}$$

Equivalently,

$$\mathcal{R}_{-a} \otimes \mathcal{R}_{-b}$$

is r-positive if and only if Equation (49) holds.

Proof. For $r \le d$, the assertion follows immediately from Theorem 4.8. If $r > d$ and $a, b \le 1/d$, both factors $I - a|\Omega\rangle\langle\Omega|$ and $I - b|\Omega\rangle\langle\Omega|$ are positive semidefinite, so their tensor product is positive semidefinite. Conversely, if, say, $a > 1/d$, choose

$$C = I_d \otimes |v\rangle\langle w|, \qquad v \perp w.$$

Then $\operatorname{rank} C = d \le r$ and $q_{a,b}(C)/\|C\|_2^2 = 1 - ad < 0$. The case $b > 1/d$ follows by interchanging the tensor factors. The map statement follows from the Choi–Jamiołkowski correspondence. □

4.4. **Quantitative trace-norm separation from bounded Schmidt number.**

Corollary 4.10 (Trace-norm separation certified by the two-copy witness)**.** *Let $1 \le r < d$ and let $Z_{r,d}$ be as in Corollary 4.3. Put*

$$\gamma := \frac{d}{r} - 1 > 0, \qquad \Delta_{r,d} := \gamma + \max\{1, \gamma^2\}.$$

Define the normalized Schmidt-number-r state set

$$\mathcal{S}_r := \{\sigma \succeq 0 : \operatorname{Tr}\sigma = 1,\ \operatorname{SN}(\sigma) \le r\}.$$

If ρ is a state satisfying

$$\operatorname{Tr}(Z_{r,d}\rho) = -\varepsilon < 0,$$

then

$$\inf_{\sigma \in \mathcal{S}_r} \|\rho - \sigma\|_1 \ge \frac{2\varepsilon}{\Delta_{r,d}}. \tag{50}$$

Proof. Since $Z_{r,d}$ is r-block-positive,

$$\operatorname{Tr}(Z_{r,d}\sigma) \ge 0 \qquad \text{for every } \sigma \in \mathcal{S}_r.$$

The operator $I - r^{-1}|\Omega\rangle\langle\Omega|$ has eigenvalues 1 and $1 - d/r = -\gamma$. Hence $Z_{r,d}$ has eigenvalues

$$1, \qquad -\gamma, \qquad \gamma^2.$$

Thus its spectral diameter is $\Delta_{r,d}$.

Fix $\sigma \in \mathcal{S}_r$. Then

$$\operatorname{Tr}\big[Z_{r,d}(\sigma - \rho)\big] \ge \varepsilon.$$

Because $\operatorname{Tr}(\sigma - \rho) = 0$, we may subtract from $Z_{r,d}$ the midpoint of its extreme eigenvalues. The resulting centered operator has operator norm $\Delta_{r,d}/2$. Hölder duality therefore gives

$$\varepsilon \le \frac{\Delta_{r,d}}{2}\|\sigma - \rho\|_1.$$

Taking the infimum over $\sigma \in \mathcal{S}_r$ proves the claim. □

4.5. **Kronecker-sum Ky Fan bounds.** By dualizing the partial-trace inequality over rank-k test operators, Theorem 1.1 yields a singular-value hierarchy for Kronecker sums.

Corollary 4.11. *Let $A, B \in M_d(\mathbb{C})$ be traceless, not necessarily Hermitian, operators, and define*

$$M := A \otimes I_d + I_d \otimes B.$$

For every $1 \le k \le d^2$,

$$\sum_{j=1}^{k} s_j(M)^2 \le \min\left\{d, k + \max\left\{0, 1 - \frac{2k}{d}\right\}\right\}\left(\|A\|_2^2 + \|B\|_2^2\right). \tag{51}$$

In particular, if $d \le 2k$, then

$$\sum_{j=1}^{k} s_j(M)^2 \le k\left(\|A\|_2^2 + \|B\|_2^2\right).$$

Proof. The Frobenius-dual form of Ky Fan duality gives

$$\left(\sum_{j=1}^{k} s_j(M)^2\right)^{1/2} = \max_{\substack{\operatorname{rank} C\le k\\ \|C\|_2=1}} |\langle C, M\rangle_{\mathrm{HS}}|. \tag{52}$$

Since A and B are traceless,

$$\begin{aligned}\langle C, M\rangle_{\mathrm{HS}} = {}& \left\langle \operatorname{Tr}_2 C - \frac{\operatorname{Tr} C}{d} I, A\right\rangle_{\mathrm{HS}} \\ &+ \left\langle \operatorname{Tr}_1 C - \frac{\operatorname{Tr} C}{d} I, B\right\rangle_{\mathrm{HS}}.\end{aligned} \tag{53}$$

By Cauchy–Schwarz,

$$\begin{aligned}|\langle C, M\rangle_{\mathrm{HS}}|^2 \le {}& \left(\|A\|_2^2 + \|B\|_2^2\right) \\ &\times \left(\|\operatorname{Tr}_1 C\|_2^2 + \|\operatorname{Tr}_2 C\|_2^2 - \frac{2}{d}|\operatorname{Tr} C|^2\right).\end{aligned} \tag{54}$$

For a rank-at-most-k operator C with $\|C\|_2 = 1$, Equation (3) gives

$$\begin{aligned}&\|\operatorname{Tr}_1 C\|_2^2 + \|\operatorname{Tr}_2 C\|_2^2 - \frac{2}{d}|\operatorname{Tr} C|^2 \\ &\le k + \left(\frac{1}{k} - \frac{2}{d}\right)|\operatorname{Tr} C|^2.\end{aligned} \tag{55}$$

If $1/k - 2/d \le 0$, the final term may be discarded. If $1/k - 2/d > 0$, then

$$|\operatorname{Tr} C|^2 \le k\|C\|_2^2 = k.$$

Thus the right-hand side of Equation (55) is at most

$$k + \max\left\{0, 1 - \frac{2k}{d}\right\}.$$

Combining this with Equations (52) and (54) gives the second entry in the minimum in Equation (51). On the other hand, tracelessness implies

$$\|M\|_2^2 = d\left(\|A\|_2^2 + \|B\|_2^2\right),$$

which gives the first entry. This proves Equation (51). □

4.6. **Exact Schmidt number and witnesses.** For $m \ge 2$, let $\Pi_-^{(m)}$ denote the orthogonal projection onto the antisymmetric subspace of $\mathbb{C}^m \otimes \mathbb{C}^m$, and let

$$\omega_-^{(m)} := \frac{\Pi_-^{(m)}}{\binom{m}{2}}$$

denote the corresponding normalized state.

The argument below invokes Lemma 2.1 directly rather than Theorem 1.1: what it needs is the bound on the top two Schmidt coefficients of a double-skew operator, not the partial-trace inequality derived from it. That $\mathrm{SN}(\omega_-^{(m)}) = 2$ is classical; the content here is the value for the tensor product.

Corollary 4.12 (Exact Schmidt number)**.** *With respect to the regrouped bipartition*

$$(A_1A_2) : (B_1B_2),$$

one has, for every $m, n \ge 2$,

$$\mathrm{SN}\left(\omega_-^{(m)} \otimes \omega_-^{(n)}\right) = 4. \tag{56}$$

Proof. Each standard antisymmetric basis vector has Schmidt rank 2, so $\omega_-^{(m)}$ admits a decomposition into pure states of Schmidt rank 2. Moreover, the antisymmetric subspace contains no nonzero product vector. Hence

$$\mathrm{SN}(\omega_-^{(m)}) = 2.$$

Tensoring Schmidt-rank-2 decompositions therefore gives

$$\mathrm{SN}\left(\omega_-^{(m)} \otimes \omega_-^{(n)}\right) \leq 4.$$

For the reverse inequality, let

$$Q := \Pi_-^{(m)} \otimes \Pi_-^{(n)},$$

where the first projection acts on A_1B_1 and the second on A_2B_2. Under inverse vectorization across $(A_1A_2) : (B_1B_2)$, every vector $\phi \in \operatorname{ran} Q$ has the form

$$\phi = \mathrm{vec}(K), \qquad K \in \mathfrak{so}(\mathbb{C}^m) \otimes \mathfrak{so}(\mathbb{C}^n).$$

Thus, for a unit vector $\phi \in \operatorname{ran} Q$, its Schmidt coefficients $s_1 \geq s_2 \geq s_3 \geq \cdots$ across the regrouped bipartition are the singular values of a Hilbert–Schmidt unit operator K in the above double-skew space. The case $m \neq n$ is included by the zero-extension argument at the beginning of Lemma 2.1. By that lemma,

$$s_1^2 + s_2^2 \leq \frac{1}{2}.$$

Since $s_3 \leq s_2 \leq s_1$,

$$2s_3^2 \leq s_1^2 + s_2^2 \leq \frac{1}{2}.$$

Consequently,

$$s_1^2 + s_2^2 + s_3^2 \leq \frac{3}{4}.$$

For any orthogonal projection Q and any k, projection duality gives

$$\begin{aligned}\sup_{\substack{\|z\|=1 \\ \mathrm{SR}(z)\leq k}} \langle z, Qz\rangle &= \sup_{\substack{\phi\in\operatorname{ran} Q \\ \|\phi\|=1}} \sup_{\substack{\|z\|=1 \\ \mathrm{SR}(z)\leq k}} |\langle z, \phi\rangle|^2 \\ &= \sup_{\substack{\phi\in\operatorname{ran} Q \\ \|\phi\|=1}} \sum_{j=1}^{k} s_j(\phi)^2;\end{aligned}$$

see [6]. Applying this identity with $k = 3$ gives

$$\sup_{\substack{\|z\|=1 \\ \mathrm{SR}(z)\leq 3}} \langle z, Qz\rangle \leq \frac{3}{4}.$$

On the other hand,

$$\omega_-^{(m)} \otimes \omega_-^{(n)} = \frac{Q}{\operatorname{rank} Q},$$

and hence

$$\mathrm{Tr}\left[Q\left(\omega_-^{(m)} \otimes \omega_-^{(n)}\right)\right] = 1.$$

If the state had Schmidt number at most 3, it would admit a convex decomposition into pure states of Schmidt rank at most 3 [7]; its expectation against Q would then be at most $3/4$, a contradiction. Therefore its Schmidt number is at least 4, proving Equation (56). □

Corollary 4.13 (Explicit Schmidt-number witnesses)**.** *Let*

$$Q = \Pi_-^{(m)} \otimes \Pi_-^{(n)}.$$

Then

$$W_2 := \frac{1}{2} I - Q$$

satisfies

$$\mathrm{Tr}(W_2 \sigma) \geq 0 \qquad \textit{whenever } \mathrm{SN}(\sigma) \leq 2,$$

and

$$W_3 := \frac{3}{4} I - Q$$

satisfies

$$\mathrm{Tr}(W_3 \sigma) \geq 0 \qquad \textit{whenever } \mathrm{SN}(\sigma) \leq 3.$$

Thus a negative expectation of W_2 certifies Schmidt number at least 3, while a negative expectation of W_3 certifies Schmidt number at least 4. For the state in Corollary 4.12,

$$\mathrm{Tr}\left[W_2 \left(\omega_-^{(m)} \otimes \omega_-^{(n)}\right)\right] = -\frac{1}{2},$$
$$\mathrm{Tr}\left[W_3 \left(\omega_-^{(m)} \otimes \omega_-^{(n)}\right)\right] = -\frac{1}{4}.$$

Appendix A. Formal verification

The proof of Theorem 1.1, and of every result it depends on, has been formalized in Lean 4. The development is available at `https://github.com/elazarg/rank-r`. It uses the toolchain `leanprover/lean4:v4.32.0`, and Mathlib is its only direct dependency, pinned in `lake-manifest.json` at revision `81a5d257c8e410db227a6665ed08f64fea08e997` (tag `v4.32.0`); that manifest also pins every transitive dependency. After `lake exe cache get`, the single command `lake build RankR` compiles it. The sources contain no `sorry`, no `axiom` declaration, no `native_decide` and no `set_option`, and `#print axioms` applied to the main theorem reports only `propext`, `Classical.choice` and `Quot.sound`; the file `RankR/Axioms.lean` wraps that check in `#guard_msgs` for each headline theorem, so it is verified by the build rather than asserted here.

The formal counterpart of Theorem 1.1 is `rank_r_partial_trace_of_FGP_square`, whose only hypotheses are $\operatorname{card} U \leq d$, $\operatorname{card} V \leq d$ and `FGPBound (Fin d) (Fin d)`, the last being the Fu–Gao–Park estimate Equation (6) on $(\mathbb{C}^d)^{\otimes 4}$. Beyond Equation (3) itself, the development covers the strict form of Equation (34), hence the assertion that for nonzero C equality in Equation (3) forces $\operatorname{rank} C = r$, and all of Remark 1.2: the extremizer, its rank, the equality it attains, and the optimality of both coefficients. Those last four are unconditional, in particular independent of the Fu–Gao–Park hypothesis. The correspondence with the results of the main text is as follows.

Manuscript	Lean
Theorem 1.1	`rank_r_partial_trace_of_FGP_square`
Lemma 2.1	`doubleSkewBound_of_FGP`
zero extension in Lemma 2.1	`FGPBound_of_card_le`
Proposition 2.2	`operatorIneq_of_doubleSkew`
Equation (10)	`Ppos_sub_Pneg`
Equation (13)	`Phi4_apply`
Equation (14)	`Phi4_ptransposeUV_rankOne`
Equation (15)	`HopScaled_eq`
Equation (20)	`Tsyn_eq_delta_vtx`
Equation (22)	`norm_Tsyn_sq_le`
Equation (23)	`inner_delta_placeQ_kron_zetaV`
Lemma 3.1	`contraction_norm`, `contraction_trace`, `hsNormSq_ptraceU_rankFactor`, `hsNormSq_ptraceV_rankFactor`
Equation (31)	`exists_rankFactor`
Equation (34)	`rank_mono`
strict case of Equation (34)	`rank_mono_strict`
Equation (2)	`rank_r_partial_trace_of_FGP _square_exact`
strictness for $\operatorname{rank} C < r$	`rank_r_partial_trace_of_FGP _square_strict`
equality forces $\operatorname{rank} C = r$	`rank_eq_of_eq_rank_r _partial_trace_of_FGP_square`
trace–rank bound in Section 3	`normSq_trace_le`
Lemma 4.7	`hsNormSq_ptraceU_le`, `hsNormSq_ptraceV_le`
Equation (46)	`ptraceU_rankOne`, `ptraceV_rankOne`
extremizer $P_r \otimes \|v\rangle\langle w\|$	`projWit`
its rank and its four norms	`rank_projWit`, `hsNormSq_projWit`, `trace_projWit`, `hsNormSq_ptraceU_projWit`, `hsNormSq_ptraceV_projWit`
equality at the extremizer	`projWit_bound_eq`
optimality of $a = r$	`le_coeff_hsNormSq_of_bound`
optimality of $b = 1/r$	`inv_le_coeff_trace_of_bound`

Three respects in which the formal statements are renderings rather than transcriptions should be recorded. First, `FGPBound` is stated in the homogeneous form Equation (6) rather than for unit vectors, and quantifies over expressions $|u_1\rangle\langle v_1| + |u_2\rangle\langle v_2|$ rather than over vectors of Schmidt rank at most two; by Equation (1) these are the same hypothesis. Second, the formal analogues of H_r, Φ and $P_\pm$ carry positive integer scalings, chosen so that no denominators occur; this does not affect positivity. Third, what is certified is the deduction, not the faithfulness of the formal definitions to the informal ones; the conventions those definitions encode are exactly the ones fixed in the introduction.

The formalization follows the arguments of Sections 2 and 3 except at two points, where it takes a shorter route: Equation (18) is replaced by a Bessel-type duality that never forms $\mathcal{T}$ as a matrix, and the complete-graph identity of Proposition B.2 is not used, the vertex Cauchy–Schwarz estimate sufficing. Those two displays are therefore not themselves covered.

Of Section 4, only Lemma 4.7 is covered, in the elementary form proved there rather than through trace-norm duality; the Schatten-1 norm appears nowhere in the development. The remaining results of Section 4, and those of Appendix B and Appendix C, are not covered. Nor are Equation (24) or Equation (35): the former is an identity between maps, which the development never forms, and the latter is what the same argument yields from a modified hypothesis that the development does not carry. Remark 2.3 is an observation about the mechanism of the formalized proof, not a further claim.

Appendix B. Gram structure and defect identities

This section records the algebraic defect structure of the rank-r proof. The main residual is represented by positive Gram operators once the range isometry is fixed; this is not claimed to be a uniform bounded-degree polynomial sum-of-squares identity in the raw Stiefel coordinates. The only literal polynomial SOS below is the elementary trace–rank residual.

For an operator C of rank at most r, define

$$\mathsf{G}_r(C) := r\|C\|_2^2 + \frac{1}{r}|\operatorname{Tr} C|^2 - \|\operatorname{Tr}_U C\|_2^2 - \|\operatorname{Tr}_V C\|_2^2, \tag{57}$$

$$\mathsf{R}_r(C) := r\|C\|_2^2 - |\operatorname{Tr} C|^2. \tag{58}$$

Proposition B.1 (Polynomial SOS for the trace–rank residual)**.** *Assume $r \le \dim(U \otimes V)$ and write*

$$C = \sum_{i=1}^{r} |e_i\rangle\langle d_i|,$$

where $e_1, \ldots, e_r$ are orthonormal and zero columns $d_i = 0$ are allowed. In $(U \otimes V) \otimes \mathbb{C}^r$, put

$$\mathbf{e} := \sum_i e_i \otimes q_i, \qquad \mathbf{d} := \sum_i d_i \otimes q_i.$$

Equip the exterior square with the convention that $u_\alpha \wedge u_\beta$ for $\alpha < \beta$ is orthonormal whenever (u_α) is an orthonormal basis. Then

$$\mathsf{R}_r(C) = \|\mathbf{e} \wedge \mathbf{d}\|^2. \tag{59}$$

Equivalently, in any orthonormal coordinate system,

$$\mathsf{R}_r(C) = \sum_{\alpha<\beta} |\mathbf{e}_\alpha \mathbf{d}_\beta - \mathbf{e}_\beta \mathbf{d}_\alpha|^2. \tag{60}$$

Proof. One has

$$\|\mathbf{e}\|^2 = r, \qquad \|\mathbf{d}\|^2 = \|C\|_2^2, \qquad |\langle \mathbf{e}, \mathbf{d}\rangle| = |\operatorname{Tr} C|.$$

The complex Lagrange identity gives both formulas. □

For the remainder of this section, fix an exact rank-r factorization

$$C = \sum_{i=1}^{r} |e_i\rangle\langle d_i|, \qquad \delta_e = \sum_i e_i \otimes q_i, \qquad \delta_d = \sum_i d_i \otimes q_i,$$

and retain ρ_0, Π_e, $P_\pm$, and Φ from the proof of Proposition 2.2. Let $\mathcal{T}_- := \mathcal{T}$ be the negative-sector synthesis map in Equation (17). Choose an orthonormal basis of $\operatorname{ran} P_+$ and let $\mathcal{T}_+$ be the synthesis map whose columns are all Kraus images under Φ of those positive-sector basis vectors. Then

$$\Phi(P_\pm) = \mathcal{T}_\pm \mathcal{T}_\pm^*.$$

For a coefficient array $c = (c_{\alpha\beta}^{(ij)})$, define

$$K_{ij} := \sum_{\alpha,\beta} c_{\alpha\beta}^{(ij)} L_\alpha \otimes M_\beta.$$

At a vertex k, associate to an incident edge the signed vectors

$$v_{k,ik} := K_{ik}\bar{e}_i \quad (i < k), \qquad v_{k,kj} := -K_{kj}\bar{e}_j \quad (k < j).$$

Proposition B.2 (Complete-graph defect certificate). *With the skew Kraus normalization used in Equation (19),*

$$\boxed{\begin{aligned}&(r-1)\|c\|_2^2-\|\mathcal{T}_-c\|^2\\ &\quad=\frac{r-1}{2}\sum_{i<j}\left[\frac{1}{2}\|K_{ij}\|_2^2-\|K_{ij}\bar{e}_i\|^2-\|K_{ij}\bar{e}_j\|^2\right]\\ &\qquad+\frac{1}{2}\sum_{k=1}^{r}\sum_{\substack{e<f\\ e,f\ni k}}\|v_{k,e}-v_{k,f}\|^2.\end{aligned}} \tag{61}$$

Consequently $\|\mathcal{T}_-\|^2\le r-1$ *and*

$$(r-1)(I-\Pi_e)-\mathcal{T}_-\mathcal{T}_-^*\succeq 0.$$

Proof. At every vertex use the exact variance identity

$$(r-1)\sum_{e\ni k}\|v_{k,e}\|^2-\|\sum_{e\ni k}v_{k,e}\|^2=\sum_{\substack{e<f\\ e,f\ni k}}\|v_{k,e}-v_{k,f}\|^2.$$

By Equations (19) and (20),

$$\begin{aligned}&(r-1)\|c\|_2^2-\|\mathcal{T}_-c\|^2\\ &\quad=\frac{r-1}{4}\sum_{i<j}\|K_{ij}\|_2^2-\frac{1}{2}\sum_{k=1}^{r}\|\sum_{e\ni k}v_{k,e}\|^2\\ &\quad=\frac{r-1}{4}\sum_{i<j}\|K_{ij}\|_2^2-\frac{r-1}{2}\sum_{i<j}\left(\|K_{ij}\bar{e}_i\|^2+\|K_{ij}\bar{e}_j\|^2\right)\\ &\qquad+\frac{1}{2}\sum_{k=1}^{r}\sum_{\substack{e<f\\ e,f\ni k}}\|v_{k,e}-v_{k,f}\|^2.\end{aligned}$$

Factoring the first two sums gives exactly Equation (61). The vertex-variance terms are nonnegative, and each edge bracket is nonnegative by Lemma 2.1. Hence $\|\mathcal{T}_-\|^2\le r-1$. Finally, Equation (23) gives $\operatorname{ran}\mathcal{T}_-\subseteq\delta_e^\perp$, whose projection is $I-\Pi_e$, so the operator inequality follows. ☐

Proposition B.3 (Exact positive/negative-sector Gram identity). *Put*

$$B_E:=(r-1)(I-\Pi_e)-\mathcal{T}_-\mathcal{T}_-^*. \tag{62}$$

Then

$$\boxed{\mathsf{G}_r(C)=\|\mathcal{T}_+^*\delta_d\|^2+\langle\delta_d,B_E\delta_d\rangle.} \tag{63}$$

Moreover, $B_E\succeq 0$.

Proof. By Lemma 3.1, $\mathsf{G}_r(C)=\langle\delta_d,H_r^{(0)}\delta_d\rangle$. Substituting Equation (15) and the two synthesis factorizations gives Equation (63). Positivity of B_E is the operator conclusion of Proposition B.2. ☐

Remark B.4 (Scope of the Gram certificate). The complete-graph identity separates the global synthesis-map defect into vertex variances and edgewise double-skew deficits. Edgewise nonnegativity is precisely the Fu–Gao–Park projection estimate used in Lemma 2.1. No uniform polynomial SOS certificate in the raw range-isometry variables is claimed.

Appendix C. Generic consequences of r-block positivity

The two results collected here are separated from Section 4 because neither uses any property of the witnesses constructed there beyond r-block positivity, and Proposition C.3 does not use Theorem 1.1 at all. They are recorded because they are what makes those witnesses usable in practice.

C.1. **Matrix-valued Schmidt-number criteria.**

Corollary C.1 (A positive-map semidefinite cut)**.** *Let $1 \le r \le d^2$, let X be a finite-dimensional Hilbert space, and put*

$$s := \min\{r, d\}, \qquad \Lambda_r := \mathcal{R}_{-1/s}^{\otimes 2}.$$

If a positive semidefinite operator $\sigma \in \mathcal{L}(X \otimes (\mathbb{C}^d)^{\otimes 2})$ has Schmidt number at most r across $X : (\mathbb{C}^d)^{\otimes 2}$, then

$$(\mathrm{id}_X \otimes \Lambda_r)(\sigma) \succeq 0. \tag{64}$$

More generally, Equation (64) holds with Λ_r replaced by $\mathcal{R}_{-a} \otimes \mathcal{R}_{-b}$ whenever $a, b \ge 0$ and

$$\max\{a, b\} \le \frac{1}{\min\{r, d\}}.$$

Proof. By Corollary 4.9, the indicated map is r-positive. Decompose σ into positive multiples of pure states of Schmidt rank at most r. For each such vector, its Schmidt support on X has dimension at most r, so positivity follows from r-positivity after restricting the identity ancilla to that support. Conjugating back into X and summing the resulting positive operators proves the assertion. □

Remark C.2 (Use as an SDP cut). Whenever a semidefinite relaxation contains an explicit candidate state block σ, Equation (64) is a linear matrix inequality in the entries of that block. If those entries are affine functions of a moment matrix, applying $\mathrm{id} \otimes \Lambda_r$ produces an affine matrix pencil and therefore a valid additional semidefinite constraint under a Schmidt-number-r promise. This is a necessary condition for that promise, not in general a complete semidefinite characterization of Schmidt number at most r.

C.2. **Local aggregation of rank witnesses.**

Proposition C.3 (Local-channel aggregation)**.** *Let $\mathcal{H}_A$ and $\mathcal{H}_B$ be finite-dimensional Hilbert spaces. For each e in a finite set E, let*

$$\Phi_e^A : \mathcal{L}(\mathcal{H}_A) \to \mathcal{L}(\mathcal{K}_{A,e}), \qquad \Phi_e^B : \mathcal{L}(\mathcal{H}_B) \to \mathcal{L}(\mathcal{K}_{B,e})$$

be quantum channels, let $W_e = W_e^* \in \mathcal{L}(\mathcal{K}_{A,e} \otimes \mathcal{K}_{B,e})$ be r-block-positive, and let $\lambda_e \ge 0$. Then

$$H := \sum_{e \in E} \lambda_e (\Phi_e^A \otimes \Phi_e^B)^*(W_e) \tag{65}$$

is r-block-positive across $\mathcal{H}_A : \mathcal{H}_B$. In particular,

$$\mathrm{Tr}(H\rho) \ge 0 \qquad \textit{whenever } \mathrm{SN}(\rho) \le r.$$

Proof. Local quantum channels do not increase Schmidt number. Indeed, write

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|, \qquad \mathrm{SR}(\psi_k) \le r,$$

and choose Kraus operators $A_{e,i}$ and $B_{e,j}$ for Φ_e^A and Φ_e^B. Then

$$(\Phi_e^A \otimes \Phi_e^B)(\rho) = \sum_{k,i,j} p_k |(A_{e,i} \otimes B_{e,j})\psi_k\rangle\langle(A_{e,i} \otimes B_{e,j})\psi_k|,$$

and every vector in this decomposition has Schmidt rank at most r. Hence

$$\mathrm{SN}\big((\Phi_e^A \otimes \Phi_e^B)(\rho)\big) \le r$$

for every e. Therefore

$$\begin{aligned}\mathrm{Tr}(H\rho) &= \sum_{e\in E} \lambda_e \, \mathrm{Tr}\left[W_e(\Phi_e^A \otimes \Phi_e^B)(\rho)\right] \\ &\ge 0.\end{aligned}$$

□

Corollary C.4 (A local-Hamiltonian Schmidt-number certificate)**.** *In Proposition C.3, suppose each local output contains two matched d_e-dimensional pairs, with $r \le d_e$, and take*

$$W_e = \left(I - \frac{1}{r}|\Omega_e\rangle\langle\Omega_e|\right)^{\otimes 2}.$$

Then the operator H in Equation (65) satisfies

$$\mathrm{Tr}(H\rho) < 0 \quad \Longrightarrow \quad \mathrm{SN}(\rho) \ge r+1.$$

For literal subsystem tests, the channels Φ_e^A and Φ_e^B may be taken to be the corresponding partial traces, so H is a sum of local terms embedded into the global Hilbert space. More generally, any geometric locality of H is inherited from the subsystem supports of the chosen channels; arbitrary channels need not produce local Hamiltonian terms.

Remark C.5 (Positive-semidefinite pulled-back terms)*.* Write

$$\Psi_e := \Phi_e^A \otimes \Phi_e^B$$

and shift the output-space witness to

$$h_e^{\mathrm{out}} := W_e - \lambda_{\min}(W_e) I_e \succeq 0,$$

where I_e is the identity on $\mathcal{K}_{A,e} \otimes \mathcal{K}_{B,e}$. Its global pullback

$$\widetilde{h}_e := \Psi_e^*(h_e^{\mathrm{out}}) \succeq 0$$

is well defined even when the output spaces depend on e. Since Ψ_e is trace-preserving, $\Psi_e^*(I_e) = I$, and hence

$$\sum_e \lambda_e \widetilde{h}_e = H - \left(\sum_e \lambda_e \lambda_{\min}(W_e)\right) I.$$

Consequently, every normalized ρ with $\mathrm{SN}(\rho) \le r$ satisfies

$$\mathrm{Tr}\left[\sum_e \lambda_e \widetilde{h}_e \, \rho\right] \ge -\sum_e \lambda_e \lambda_{\min}(W_e).$$

For the witnesses in Corollary C.4,

$$\lambda_{\min}(W_e) = 1 - \frac{d_e}{r},$$

so the known energy baseline is

$$\sum_e \lambda_e \left(\frac{d_e}{r} - 1\right).$$

This construction certifies Schmidt number across the prescribed bipartition; by itself it is not an NLTS statement.

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